# DR-35567

## Design review report

**DR-35567 Rev 1**
2026-04-08
Released
Confidential

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|---|---|
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Mod. 501 02/2025 Rev.3

# 1 Summary

Aim of this document is to describe suggestions and corrections that Telit advises to improve Ovoo Electronics customer application that integrates a Telit-Cinterion LE910C1-EUX module.

Design review is based on the following received documentation:

- Schematic files: B101BA_00.00.01_Schematic.pdf
- Gerber file: Gerber folder
- Other: Telit Design Review 01 (Schematic & PCB).pdf, B101BA_00.00.01_BOM.csv

Summary Tables:

| Schematic Review | OK | Error | Warning | Tip | Missing Info | Not Applicable |
|---|---|---|---|---|---|---|
| Power Supply | | | △ | ▢ | | |
| SIM Pins | ○ | | | | | |
| Digital Pins | | ✖ | △ | | | |
| RF Aspects | | | △ | | | |
| Audio Aspects | | | | | | ⌀ |

| PCB Layout Review | OK | Error | Warning | Tip | Missing Info | Not Applicable |
|---|---|---|---|---|---|---|
| General Placement | | ✖ | | | | |
| Antenna Waveguide | ○ | | | | | |
| RF Aspects | ○ | | | | | |
| Audio Aspects | | | | | | ⌀ |

The following symbols will be used throughout the Design Review to indicate:

- ○ OK: No design changes are required.
- ▢ Tip: information or possible improvement, not mandatory but recommended.
- △ Warning: if you don't follow the recommendation there's a risk of malfunctioning or issues during the certification phase, strongly recommended.
- ✖ Error: it's mandatory to follow the recommendation, otherwise the module could be damaged or could not work properly or there are high chances of facing issues during the certification/approval phase.
- ◌ Missing Information: some relevant information is missing therefore the DR cannot be accurate on this item.
- ⌀ Not Applicable: the feature does not apply to the design under review.

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# 2 Design Review

## 2.1 Schematic review

### 2.1.1 Power supply

- ⚠ Since the application will be close to a high magnetic field, we strongly suggest changing the inductor L3 with one that is shielded to avoid its saturation.
- ○ In system where there is the possibility of a sudden power cut-off, we suggest implementing the fast-shutdown feature to avoid the memory corruption of the modem. For more information, please refer to the chapter FAST_SHUTDOWN in the Hardware User Guide.

### 2.1.2 Sim pins

- ○ O.K.

### 2.1.3 Digital pins

#### 2.1.3.1 Control signals

- ○ O.K.

#### 2.1.3.2 Communication interfaces

- ⚠ To avoid module latch-up, we suggest using the VAUX/PWRMON signal on VCCB line of the voltage translator instead of VIO_1V8.

#### 2.1.3.3 Other

- ✖ The pins K4 and M6 are reserved and mustn't be connected.

### 2.1.4 RF aspects

- ⚠ To avoid harmonics, it's suggested to predisposing, on modem board and in any other board connected or close to it, 33pF capacitors to GND, placed very close to the component pads we want to filter and on its same path and layer. See examples on where the use of this filtering is very advised:

  o Power Sources and signals on input-output connectors

  o On power supply output PADs

  o On component's power supply input PADs (even if shielded)

  o Diodes in forward conduction, like LEDs, on anode and/or cathodes if not directly tied to a power net (LED D6)

o Transistor bases, mainly for bipolar ones, phototransistors and opto-isolator ()

The capacitor value 33pF is suggested because at this value, in general, the SMD 0603/0402, 25V/50V COG capacitors have self-resonance at about 1GHz. For 0201 components the 1GHz resonance is on values of 10pF to 20pF.

### 2.1.5 Audio pins

- N.A.

## 2.2 PCB Layout review

### 2.2.1 General placement

- ✖ The footprint of the modem is wrong: below is depicted the correct footprint.

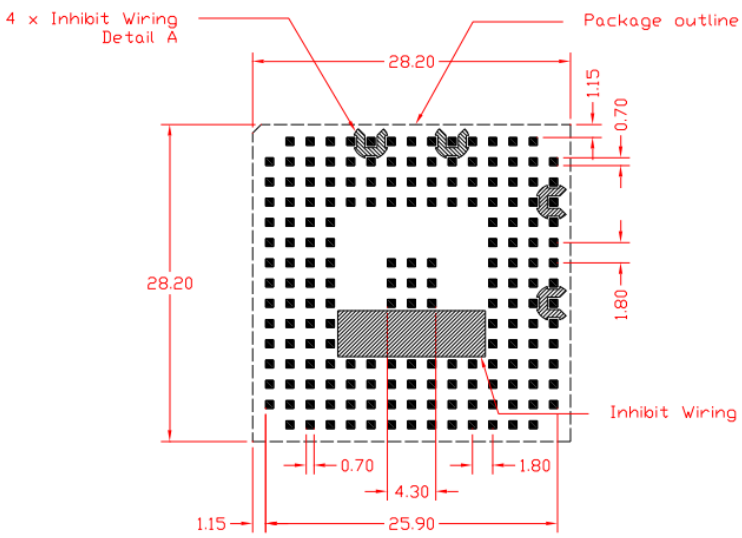


Figure 1. Modem footprint recommended.

### 2.2.2 RF aspects

- ○ The RF waveguide has the 50 Ohm characteristic impedance required.

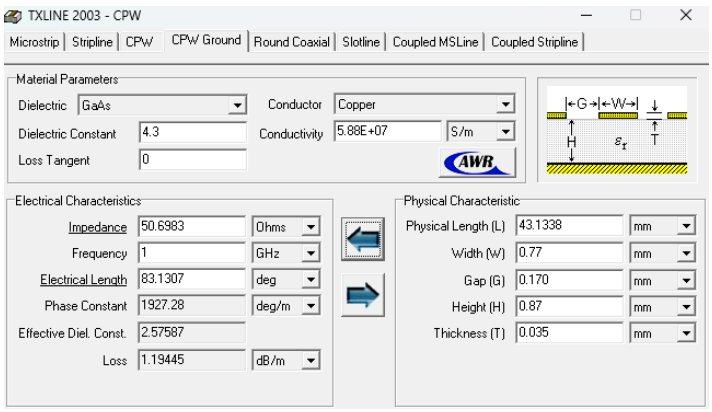


Figure 2. RF waveguide parameters calculated.

### 2.2.3 Audio aspects

- N.A.

## 2.3 General comments

Review is related to received application information and the supposed use of it.

# 3 Quality record

Please, take a couple of minutes to fill out our feedback form by clicking on the following link:

Telit Customer satisfaction survey link

This design review is registered internally in JIRA with ID #DR-35567.

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